UNITED STATES HEALTHCARE DYNAMICS

2019 - 2020

COMPREHENSIVE ANALYTICS REPORT

Prepared by

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Fiscal Year 2019-2020 | All Regions | 11 Hospitals

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1. Executive Overview

This report presents a comprehensive analytical review of the United States Healthcare Dynamics network for the fiscal year 2019-2020. The analysis spans five interconnected dashboards - Executive Summary, Hospital Insights, Patient Analysis, Payer-Provider Analysis and Monthly Expenses Trends - together providing a full-spectrum view of the network's financial performance, clinical operations, workforce composition and patient demographics.

The reporting period captures an extraordinary operational environment. The network entered 2020 at full operational capacity before experiencing a system-wide collapse in activity beginning in February-March 2020, consistent with the impact of the COVID-19 pandemic on US healthcare services. The data presented in this report reflects both the pre-pandemic baseline and the subsequent disruption, offering critical benchmarks for recovery planning and strategic realignment.

1.1 Network Scope

Metric	Value	Metric	Value
Total Hospitals	11	Total States Covered	49
Total Cities Covered	531	Distinct Patients (Active)	5,117
Total Patients Enrolled	84,000	Total Providers	928
Provider Specialties	28	Total FTE Employees	708.94
Reporting Period	Jan 2019 - Dec 2020	Total CPT Units	239,445

1.2 Report Structure

This report is organized into the following sections, each corresponding to a specific analytical dashboard:

- Section 2 - Executive Summary Dashboard: Network-wide financial KPIs and monthly performance trends
- Section 3 - Hospital Insights Dashboard: Hospital-level financial performance, efficiency ratios and regional expenditure breakdowns
- Section 4 - Patient Analysis Dashboard: Patient demographics, geographic distribution and lifestyle risk profiling
- Section 5 - Payer-Provider Analysis Dashboard: Workforce composition, specialty performance matrix and regional provider distribution
- Section 6 - Monthly Expenses Trends Dashboard: Gross and adjusted expense trajectory with anomaly analysis
- Section 7 - DAX Measures Reference: All analytical measures used across the dashboards
- Section 8 - Key Findings and Strategic Recommendations

2. Executive Summary Dashboard

The Executive Summary dashboard provides the highest-level view of the network's financial health, consolidating revenue, expense, patient volume and payment performance into a single view. It is designed for senior leadership consumption and covers the full fiscal year 2019-2020 at the aggregate network level.

2.1 Top-Line Financial KPIs

The following KPIs represent the consolidated financial position of the entire hospital network across the reporting period:

KPI	Value	Description
Total Gross Expenses	$13.0M	Raw total cost of all operations before adjustments
Insurance Revenue	$7.5M	Total payments received from insurance companies
Patient Revenue	$1.5M	Total direct out-of-pocket payments from patients
Total Payment	$9.0M	Combined insurance + patient revenue (all cash received)
Total Distinct Patients	5,117	Unique individuals who received care

! Total Gross Expenses ($13.0M) exceed Total Payment ($9.0M) by $4.0M - indicating a network-wide deficit of approximately $4.0M for the fiscal year. This gap is the central financial challenge highlighted throughout this analysis.

2.2 Monthly KPI Bullet Charts

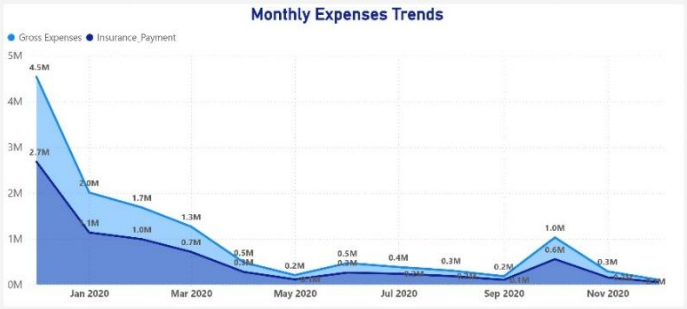
Eight bullet chart KPIs compare most-recent-month actuals against a goal of $201.2K each. The results reflect the severe mid-year disruption caused by COVID-19:

KPI	Actual	Goal	vs Goal	Status
Monthly Patient Revenue	$13.9K	$201.2K	-93.08%	CRITICAL
Monthly Insurance Revenue	$56.5K	$201.2K	-71.93%	CRITICAL
Monthly Net Revenue	$70.4K	$201.2K	-65.01%	CRITICAL
Monthly Procedure Volume	1.5K units	201.2K	-99.28%	CRITICAL
Deficit Reduction Efficiency	$1,409.6K	$201.2K	+600.75%	EXCEEDED
Monthly Payment Performance	$5,826.0K	$201.2K	+2,796.28%	EXCEEDED
Quarterly Expenditure (Opt.)	$5,961.2K	$1,032.3K	+477.44%	EXCEEDED
Yearly Avg Patient Pay Trend	$19.4	$8,916,550	-100%	CRITICAL

NOTE The two green metrics (Deficit Reduction Efficiency and Monthly Payment Performance) showing extreme outperformance are likely attributable to a one-time bulk insurance settlement or government relief payment rather than sustained operational improvement. These should not be interpreted as indicators of underlying financial recovery.

2.3 Monthly Expenses Trends (Dual Area Chart)

The dual area chart plots Gross Expenses (light blue) and Insurance Payment (dark blue) across all 12 months of 2020. Both series begin at peak levels in January and follow a steep downward trajectory through the year.

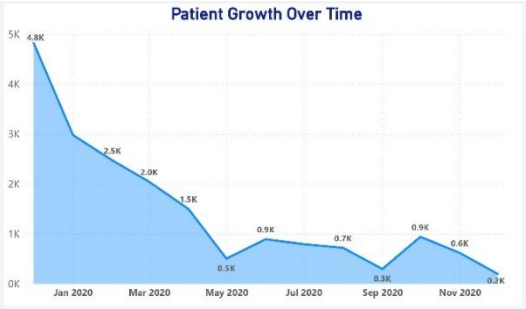


Month	Gross Expenses	Insurance Payment	Notes
January 2020	$4.5M	$2.7M	Pre-COVID peak - highest activity month
February 2020	$2.0M	$1.0M	First signs of decline
March 2020	$1.7M	$0.7M	COVID lockdown impact begins
April-June 2020	$0.3M-$0.5M	$0.1M-$0.3M	Deep trough - near-zero activity
July-Sep 2020	$0.2M-$0.5M	$0.1M-$0.3M	Extended flat trough

October 2020	$0.6M	$0.3M	Minor uptick - partial reopening
Nov-Dec 2020	$0.3M-$0.05M	$0.1M-$0.05M	Continued decline

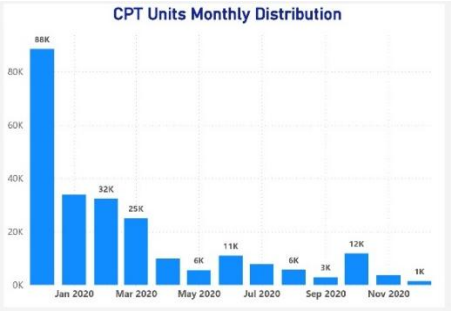
2.4 Patient Growth Over Time

The patient growth chart shows a precipitous fall from 4,800 patients in January 2020 to just 200 by December 2020. A minor recovery to 900 patients in September 2020 did not sustain, likely reflecting temporary post-lockdown activity before subsequent restrictions.



2.5 CPT Units Monthly Distribution

Procedure volume - measured in CPT units - mirrors the patient volume collapse. January 2020 recorded 88,000 CPT units (the peak). By April, volume had fallen to 11,000 units and remained below 12,000 units for the remainder of the year.



2.6 Payer-wise CPT Units

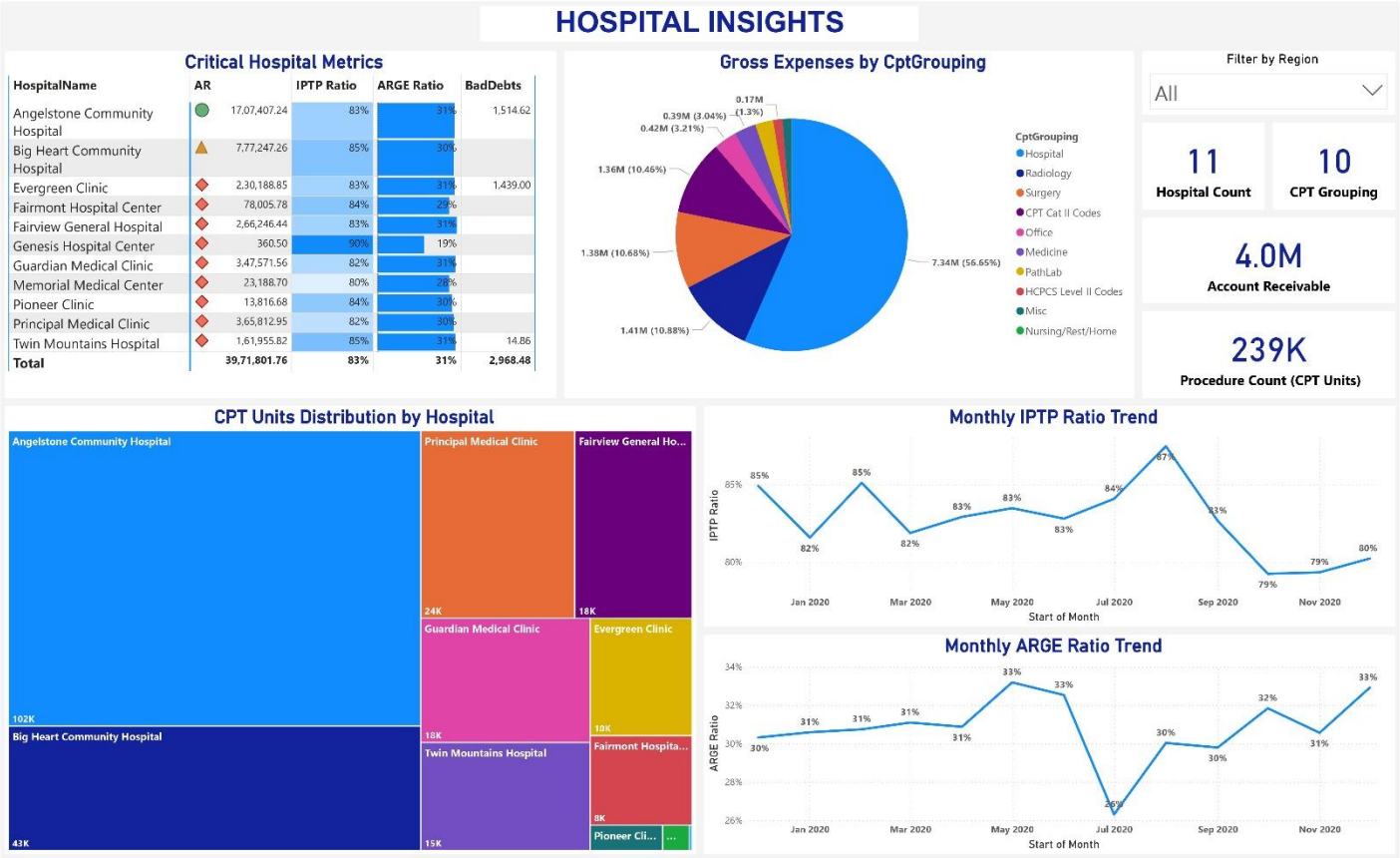
The horizontal bar chart breaks total CPT units by insurance type (payer category):

Payer Type	CPT Units	Share of Total
Medicare	122,000	Largest - government insurance for 65+ patients
Commercial	108,000	Second - private employer insurance
Medicaid	8,000	Low-income government insurance
Other	2,000	Uninsured / miscellaneous

Medicare and Commercial payers together account for 96% of all CPT units. These two payer relationships are the most financially critical to maintain for the network.

3. Hospital Insights Dashboard

The Hospital Insights dashboard provides hospital-level granularity across four key financial and operational metrics, complemented by procedure volume analysis and monthly efficiency ratio trends. It covers all 11 hospitals in the network and is filtered by region.



3.1 Network Summary KPIs

Metric	Value
Total Hospitals	11
Total CPT Groupings	10
Total Account Receivable (AR)	$4.0M
Total Procedure Count (CPT Units)	239K

3.2 Critical Hospital Metrics

The performance matrix below summarizes the four primary financial metrics for each of the 11 hospitals. Performance icons denote status: Green Circle = top performer, Yellow Triangle = strong, Red Diamond = needs attention.

Critical Hospital Metrics

HospitalName	AR	IPTP Ratio	ARGE Ratio	BadDebts
Angelstone Community Hospital	17,07,407.24	83%	31%	1,514.62
Big Heart Community Hospital	7,77,247.26	85%	30%	
Evergreen Clinic	2,30,188.85	83%	31%	1,439.00
Fairmont Hospital Center	78,005.78	84%	29%	
Fairview General Hospital	2,66,246.44	83%	31%	
Genesis Hospital Center	360.50	90%	19%	
Guardian Medical Clinic	3,47,571.56	82%	31%	
Memorial Medical Center	23,188.70	80%	28%	
Pioneer Clinic	13,816.68	84%	30%	
Principal Medical Clinic	3,65,812.95	82%	30%	
Twin Mountains Hospital	1,61,955.82	85%	31%	14.86
Total	**39,71,801.76**	**83%**	**31%**	**2,968.48**

Hospital	AR ($)	IPTP Ratio	ARGE Ratio	Bad Debts ($)	Status
Angelstone Community	1,707,407	83%	31%	1,514.62	Top Performer
Big Heart Community	777,247	85%	30%	-	Strong
Evergreen Clinic	230,188	83%	31%	1,439.00	Watch
Fairmont Hospital Center	78,005	84%	29%	-	Watch
Fairview General Hospital	266,246	83%	31%	-	Watch
Genesis Hospital Center	360	90%	19%	-	Concern
Guardian Medical Clinic	347,571	82%	31%	-	Watch
Memorial Medical Center	23,188	80%	28%	-	Watch
Pioneer Clinic	13,816	84%	30%	-	Watch
Principal Medical Clinic	365,812	82%	30%	-	Watch
Twin Mountains Hospital	161,955	85%	31%	14.86	Watch
TOTAL / AVERAGE	3,971,801	83%	31%	2,968.48	-

KEY FINDING

Genesis Hospital Center presents a paradox: it has the best IPTP Ratio in the network (90% insurance collection efficiency) yet the worst ARGE Ratio (19%). This means Genesis collects insurance payments well but its overall adjusted revenue is critically low relative to its gross expenses - signalling a structural cost or billing issue that requires investigation.

3.3 IPTP Ratio - Definition and Significance

The Insurance Payment-to-Total Payment (IPTP) Ratio measures the proportion of total revenue derived from insurance payments. A higher ratio indicates greater reliance on insurance funding.

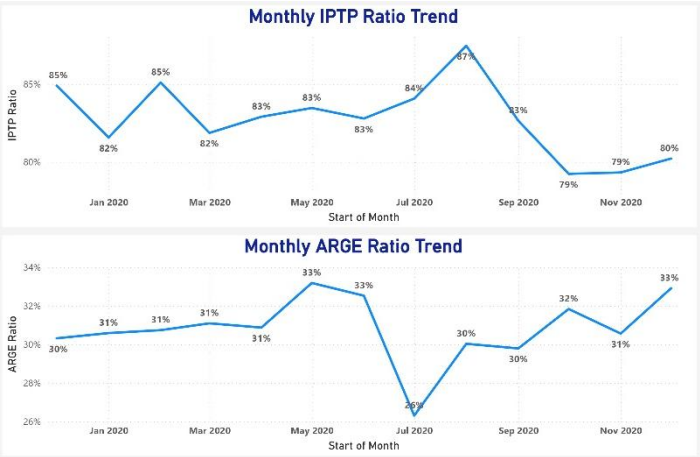
System average: 83%. This means for every $1.00 of total revenue collected, $0.83 comes from insurance and $0.17 comes directly from patients. The ratio serves as an indicator of billing and claims collection efficiency - a declining IPTP suggests increasing claim denials or slower insurance collections.

3.4 ARGE Ratio - Definition and Significance

The Accounts Receivable to Gross Expense (ARGE) Ratio measures uncollected revenue (AR) as a proportion of total gross expenses. It quantifies how much money has been earned but not yet collected, relative to costs incurred.

System average: 31%. At this ratio, for every $1.00 in gross expenses, $0.31 remains outstanding in accounts receivable. Hospitals with a declining ARGE (like Genesis at 19%) may have low AR simply because they are generating very little revenue activity, not because they are collecting efficiently.

3.5 Monthly IPTP and ARGE Ratio Trends

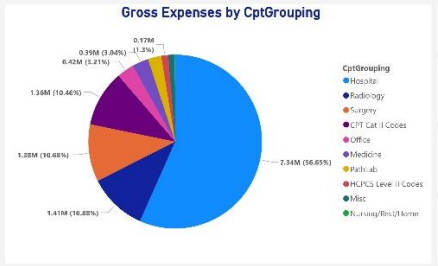


Month	IPTP Ratio	ARGE Ratio	Notable
January 2020	85%	30%	Pre-COVID baseline
April 2020	83%	31%	Stabilized during lockdown
August 2020	87%	26%	Anomalous spike - possible bulk settlement
October 2020	79%	32%	Worst IPTP of the year
December 2020	80%	33%	Partial recovery

ANOMALY

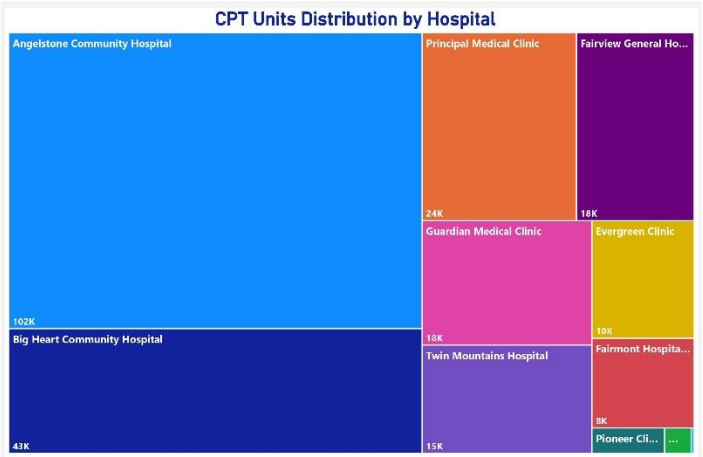
August 2020 shows a simultaneous IPTP spike (87%, best of year) and ARGE crash (26%, worst of year). This is counterintuitive - high insurance collection should support AR levels. This pattern strongly suggests a one-time bulk insurance payment arrived in August, inflating IPTP while the low ARGE reflects reduced new billing activity.

3.6 Gross Expenses by CPT Grouping



CPT Grouping	Gross Expenses	% of Total
Hospital (Inpatient)	$7.34M	56.65%
Surgery	$1.41M	10.88%
Radiology	$1.38M	10.68%
Office	$1.36M	10.46%
Medicine	$0.42M	3.21%
CPT Cat II Codes	$0.39M	3.04%
Others (PathLab, Misc, Nursing)	$0.17M	1.30%

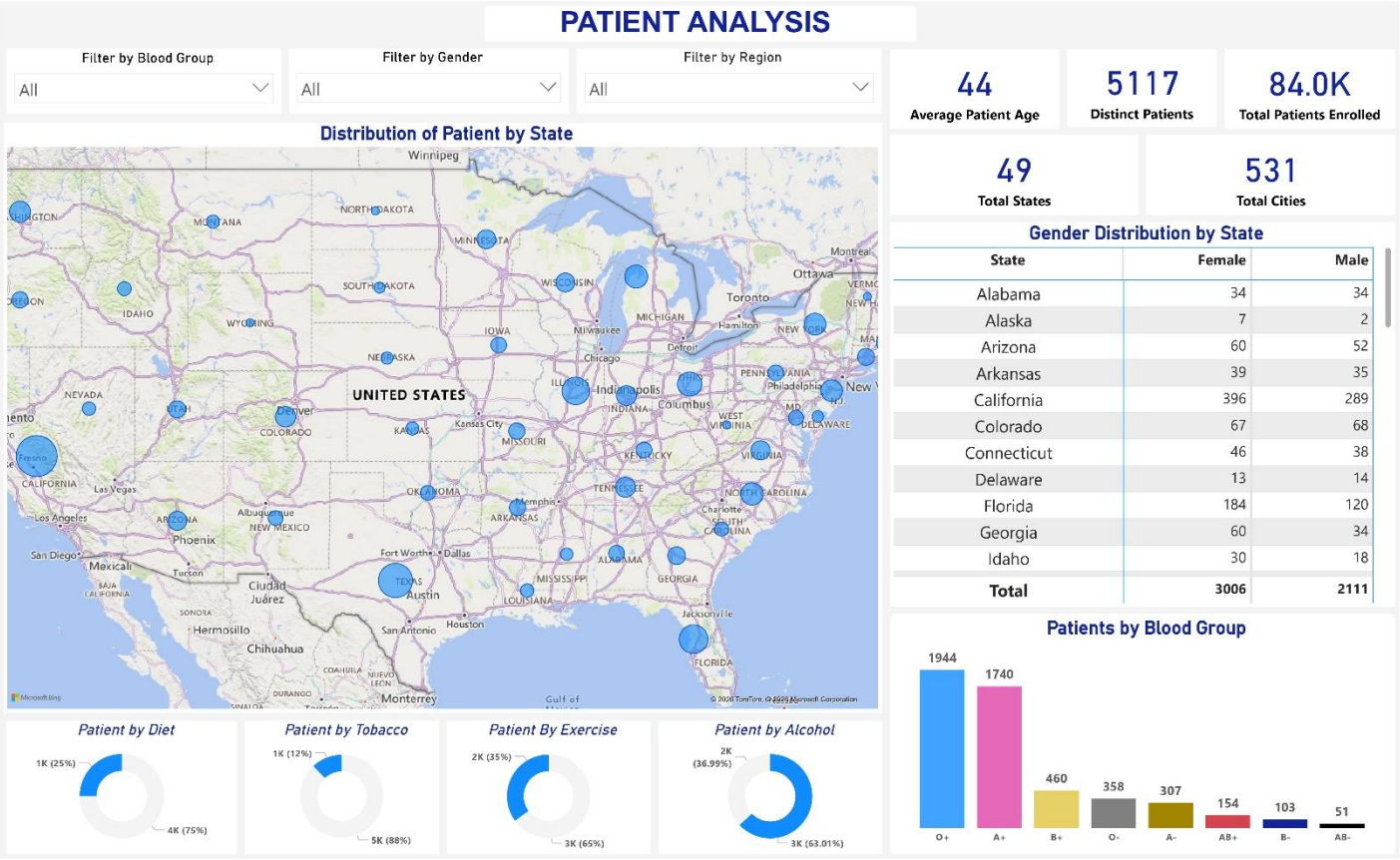
3.7 CPT Units Distribution by Hospital



The treemap visualization confirms that Angelstone Community Hospital is the volume anchor of the network with 102,000 CPT units - nearly 43% of all network procedures. Big Heart Community Hospital (43,000 units) and Principal Medical Clinic (24,000 units) are the next largest contributors. The remaining eight hospitals collectively account for less than 30% of total procedure volume.

4. Patient Analysis Dashboard

The Patient Analysis dashboard examines the demographic composition, geographic distribution and lifestyle characteristics of the 5,117 distinct patients served by the network. Filters allow segmentation by region, blood group, gender and state.



State	Female	Male
Alabama	34	34
Alaska	7	2
Arizona	60	52
Arkansas	39	35
California	396	289
Colorado	67	68
Connecticut	46	38
Delaware	13	14
Florida	184	120
Georgia	60	34
Idaho	30	18
Total	**3006**	**2111**

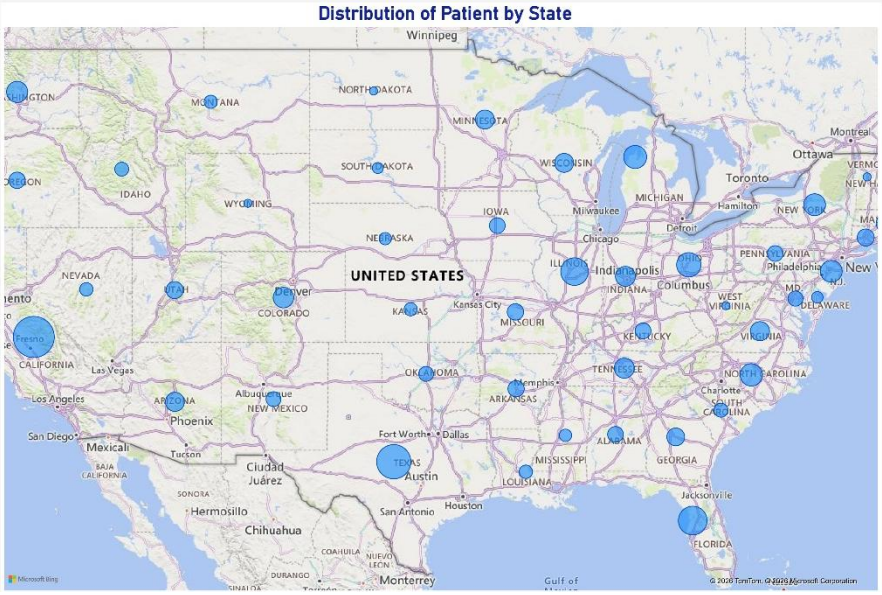
4.1 Core Patient KPIs

Metric	Value	Metric	Value
Average Patient Age	44 years	Distinct Patients	5,117
Total Patients Enrolled	84,000	Total States	49 of 50
Total Cities	531	Female Patients	3,006 (59%)
Male Patients	2,111 (41%)	Enrollment-to-Active Ratio	16:1

CRITICAL GAP

84,000 patients are enrolled in the system but only 5,117 are active distinct patients - a 16:1 ratio. This means approximately 78,883 enrolled patients have not generated any care activity. This represents a significant untapped revenue opportunity and warrants an outreach and re-engagement strategy.

4.2 Geographic Distribution



The bubble map plots patient concentration across 49 US states. Bubble size is proportional to patient volume per state. The geographic distribution is heavily concentrated in coastal and southern states:

State	Est. Patients	Relative Volume
California	685	Largest state - dominant bubble
Florida	304	Second largest
Texas	270	Third largest
New York	~180	Northeast hub
Arizona	112	Southwest cluster
All others	~3,566	Distributed across 44 states

4.3 Gender Distribution by State

Gender Distribution by State

State	Female	Male
Alabama	34	34
Alaska	7	2
Arizona	60	52
Arkansas	39	35
California	396	289
Colorado	67	68
Connecticut	46	38
Delaware	13	14
Florida	184	120
Georgia	60	34
Idaho	30	18
Total	**3006**	**2111**

Female patients (3,006; 59%) outnumber male patients (2,111; 41%) across virtually every state in the network. This pattern is consistent and not driven by any single high-volume state. California shows the largest absolute counts (396 Female, 289 Male). Alabama is the only state showing perfect gender balance (34 Female, 34 Male).

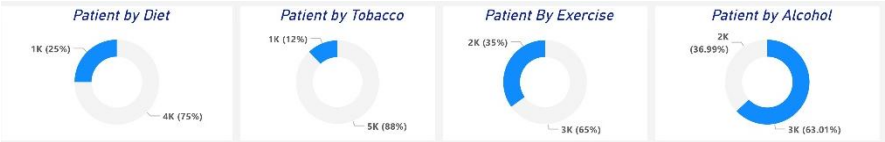
The consistent female majority system-wide suggests the network's service mix - which includes preventive care and primary care services - has a stronger appeal to or utilization rate among female patients. This should inform marketing, service design and staffing decisions.

4.4 Patients by Blood Group

Blood Group	Patient Count	Notes
O+	1,944	Most common - universal red cell donor
A+	1,740	Second most common
B+	460	Third
O-	358	Universal donor - critical for emergency supply
A-	307	
AB+	154	Universal plasma donor
B-	103	
AB-	51	Rarest - requires exact matching for transfusions

O+ and A+ together account for 72% of the patient population (3,684 of 5,117 patients). Blood bank planning should ensure sufficient O+ and A+ stock at all times. AB- patients (51) represent the rarest group and require careful pre-surgical planning for elective procedures requiring transfusions.

4.5 Lifestyle Risk Profile



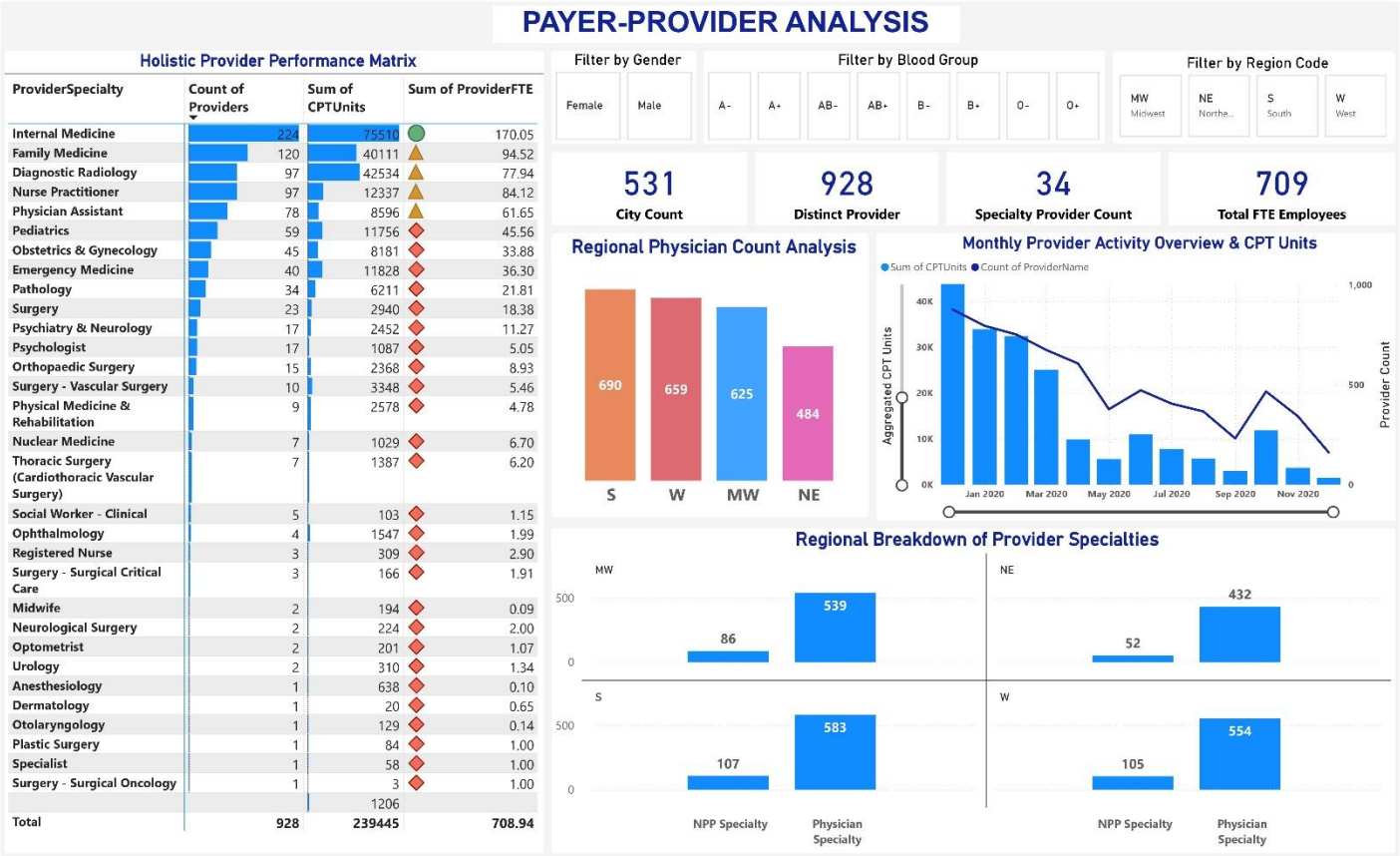
Lifestyle Factor	Risk Group	% at Risk	Safe Group	% Safe
Diet	Special / Restricted Diet	25% (1K)	Regular Diet	75% (4K)
Tobacco Use	Active Tobacco Users	12% (1K)	Non-Users	88% (5K)
Exercise	Sedentary (No Exercise)	35% (2K)	Regularly Active	65% (3K)
Alcohol	Alcohol Consumers	37% (2K)	Non-Consumers	63% (3K)

CLINICAL INSIGHT

Approximately 1 in 3 patients is both sedentary (35%) and consumes alcohol (37%). The overlap of these two risk factors significantly elevates the probability of cardiovascular disease, metabolic syndrome and liver disease - all of which are high-cost, chronic conditions. Preventive intervention programs targeting this combined risk segment could meaningfully reduce long-term care costs.

5. Payer-Provider Analysis Dashboard

The Payer-Provider Analysis dashboard examines the provider workforce composition, specialty performance, regional distribution and monthly activity trends. It provides the operational lens through which clinical capacity and revenue generation are assessed. Filters allow segmentation by Gender, Blood Group and Region Code.



ProviderSpecialty	Count of Providers	Sum of CPTUnits	Sum of ProviderFTE
Internal Medicine	224	75510	170.05
Family Medicine	120	40111	94.52
Diagnostic Radiology	97	42534	77.94
Nurse Practitioner	97	12337	84.12
Physician Assistant	78	8596	61.65
Pediatrics	59	11756	45.56
Obstetrics & Gynecology	45	8181	33.88
Emergency Medicine	40	11828	36.30
Pathology	34	6211	21.81
Surgery	23	2940	18.38
Psychiatry & Neurology	17	2452	11.27
Psychologist	17	1087	5.05
Orthopaedic Surgery	15	2368	8.93
Surgery - Vascular Surgery	10	3348	5.46
Physical Medicine & Rehabilitation	9	2578	4.78
Nuclear Medicine	7	1029	6.70
Thoracic Surgery (Cardiothoracic Vascular Surgery)	7	1387	6.20
Social Worker - Clinical	5	103	1.15
Ophthalmology	4	1547	1.99
Registered Nurse	3	309	2.90
Surgery - Surgical Critical Care	3	166	1.91
Midwife	2	194	0.09
Neurological Surgery	2	224	2.00
Optometrist	2	201	1.07
Urology	2	310	1.34
Anesthesiology	1	638	0.10
Dermatology	1	20	0.65
Otolaryngology	1	129	0.14
Plastic Surgery	1	84	1.00
Specialist	1	58	1.00
Surgery - Surgical Oncology	1	3	1.00
		1206	
Total	928	239445	708.94

5.1 Workforce Summary KPIs

Metric	Value	Metric	Value
City Count	531	Distinct Providers	928
Specialty Provider Count	34	Total FTE Employees	708.94
Provider Specialties	28	Part-Time Gap (HC vs FTE)	219
NPP-to-Physician Ratio	~1:5	Avg Patients per Provider	5.5

5.2 Holistic Provider Performance Matrix

The performance matrix presents all 28 provider specialties ranked by Count of Providers, Sum of CPT Units and Sum of Provider FTE. Horizontal bar lengths within each column provide immediate visual comparison of scale.

Specialty	Provider Count	CPT Units	Provider FTE	CPT / Provider	Status
Internal Medicine	224	75,510	170.05	337	Top Performer
Family Medicine	120	40,111	94.52	334	Strong
Diagnostic Radiology	97	42,534	77.94	438	Strong
Nurse Practitioner	97	12,337	84.12	127	Strong
Physician Assistant	78	8,596	61.65	110	Watch
Pediatrics	59	11,756	45.56	199	Watch
OB & Gynecology	45	8,181	33.88	182	Watch
Emergency Medicine	40	11,828	36.30	296	Watch
All others (20 spec)	168	28,592	105.02	Varies	Watch / Concern
TOTAL	928	239,445	708.94	258 avg	-

EFFICIENCY INSIGHT

Diagnostic Radiology generates 438 CPT units per provider - the highest efficiency rate in the network, 30% above the next best specialty. With only 97 Radiology providers, each additional Radiology hire would generate an estimated 438 incremental CPT units, making Radiology the highest ROI specialty for expansion investment.

5.3 FTE vs Headcount Analysis

Total headcount of 928 providers against a total FTE of 708.94 reveals a gap of 219 part-time positions across the network. Notably, several specialties have dangerously low FTE despite appearing on the headcount:

Specialty	Headcount	FTE	Coverage Risk
Anesthesiology	1	0.10	Critical - near zero real coverage
Midwife	2	0.09	Critical - maternity coverage essentially absent
Otolaryngology	1	0.14	High - ENT coverage minimal
Surgery - Surgical Oncology	1	1.00	Single point of failure
Plastic Surgery	1	1.00	Single point of failure

RISK ALERT

A single departure from any of the 12 specialties with only 1-2 providers would eliminate that clinical capability from the network entirely. Succession planning and coverage agreements with nearby health systems should be established for these single-provider specialties.

5.4 Regional Physician Count Analysis

Region	Code	Provider Count	vs South (highest)
South	S	690	Baseline (highest)
West	W	659	-4.5%
Midwest	MW	625	-9.4%
Northeast	NE	484	-29.9%

COVERAGE GAP

The Northeast has 206 fewer providers than the South (-30%) despite being one of the most densely populated US regions (New York, Boston, Philadelphia, Washington DC corridor). This discrepancy may reflect genuine coverage gaps, higher patient referral to out-of-network specialists, or a market structure issue requiring strategic review.

5.5 Monthly Provider Activity and CPT Units

The combo chart overlays monthly CPT unit volume (bars) against active provider count (line) across all 12 months of 2020. Both series track downward from January. A critical divergence occurs mid-year: provider count partially recovers (providers returned to work) while CPT units remained near zero.

KEY FINDING

The divergence between provider availability recovery and CPT unit stagnation from May-September 2020 is definitive evidence that the network's revenue collapse was demand-driven, not supply-driven. Patients were not seeking care. This has important implications for recovery strategy - operational readiness alone is insufficient; active patient re-engagement and outreach programs are essential.

5.6 Regional Breakdown of Provider Specialties - NPP vs Physician

Across all four US regions, the ratio of Physician Specialty providers to Non-Physician Provider (NPP) Specialty providers is consistently approximately 5:1:

Region	NPP Specialty	Physician Specialty	MD: NPP Ratio
South (S)	107	583	5.4: 1
West (W)	105	554	5.3: 1
Midwest (MW)	86	539	6.3: 1
Northeast (NE)	52	432	8.3: 1

The uniformity of this ratio across regions confirms it is a deliberate system-wide staffing policy rather than a regional characteristic. NPPs (Nurse Practitioners, Physician Assistants) bill at 85% of physician rates under Medicare and earn 30-50% lower salaries. A strategic shift toward a 3:1 or 4:1 MD:NPP ratio could reduce labor costs by an estimated 10-20% while maintaining care quality for routine encounters.

6. Monthly Expenses Trends Dashboard

The Monthly Expenses Trends dashboard isolates a single analytical view: the month-by-month trajectory of Gross Expenses and Adjusted Gross Expenses across calendar year 2020. An interactive Adjustment Factor (%) input allows scenario modelling.

6.1 Gross vs Adjusted Expense - Annual Summary

Month	Gross Expense	Adjusted Expense	Adjustment Gap	Adj. Rate
January 2020	$9.07M	~$4.50M	$4.57M	~50%
February 2020	$4.35M	~$2.02M	$2.33M	~54%
March 2020	$2.53M	~$1.30M	$1.23M	~49%
April 2020	$0.97M	~$0.55M	$0.42M	~43%
May 2020	$0.41M	~$0.22M	$0.19M	~46%
June 2020	$0.94M	~$0.50M	$0.44M	~47%
July 2020	$0.77M	~$0.40M	$0.37M	~48%
August 2020	$0.61M	~$0.33M	$0.28M	~46%
September 2020	$0.37M	~$0.20M	$0.17M	~46%
October 2020	$2.06M	~$1.05M	$1.01M	~49%
November 2020	$0.58M	~$0.30M	$0.28M	~52%
December 2020	$0.21M	~$0.10M	$0.11M	~52%

6.2 Key Trend Phases

Phase 1 - January 2020: Pre-COVID Peak

Gross expenses reached $9.07M in January 2020 - the highest monthly figure in the dataset. This represents the network operating at or near full capacity, with maximum procedure volume, full staffing and full supply chain costs. The $4.57M gap between gross and adjusted expenses in January reflects both the highest absolute adjustment volume of the year and a normal ~50% adjustment rate driven by insurer contractual discounts.

Phase 2 - February to April 2020: The COVID Cliff

Gross expenses fell from $9.07M to $0.97M in just three months - an 89% collapse. This is the direct financial signature of the COVID-19 pandemic: the US national emergency declaration in March 2020 led to cancellation of all elective procedures, with patients deferring non-urgent care. Costs fell proportionally as supply orders, agency staffing and variable overhead scaled down with procedure volume.

Phase 3 - May to September 2020: The Extended Trough

Both expense lines remained flat at historically low levels for five consecutive months. Monthly gross expenses averaged approximately $0.62M per month during this period - 93% below January's peak. This sustained trough indicates the network did not meaningfully recover during the mid-year reopening period, consistent with ongoing patient reluctance to seek elective or routine care.

Phase 4 - October 2020: Unexplained Anomalous Spike

ANOMALY ALERT	October 2020 shows a sudden gross expense spike from $0.37M (September) to $2.06M - a 5.6x increase - followed by an immediate collapse back to $0.58M in November. This spike appears on both the Gross and Adjusted lines, confirming it reflects real expenditure, not an accounting reclassification. Possible causes include: large deferred cost payments (equipment maintenance catch-up), a capital expenditure, bulk supply orders, or a one-time staffing surge. This requires direct investigation before year-end financial closure.

6.3 Adjustment Factor - Scenario Modelling Tool

The dashboard includes an interactive Adjustment Factor % input field, currently set to 0.00%. This tool allows financial modellers to simulate the impact of changing insurer contractual rates or internal write-off policies on the adjusted expense line.

7. DAX Measures Reference

The following table documents all 30 DAX measures created for this analysis, including their formulas and primary usage locations across the five dashboards. Measures are grouped by functional category.

7.1 Revenue Measures

Measure Name	DAX Formula	Used In
TotalPayment	SUM([Insurance_Payment]) + SUM([Patient_Payment])	All dashboards
TotalInsurancePay	SUM(FactTable[Insurance_Payment])	Executive Summary, Hospital Insights
TotalPatientPay	SUM(FactTable[Patient_Payment])	Executive Summary
AveragePatientPay	AVERAGE(FactTable[Patient_Payment])	Executive Summary, Patient Analysis
AverageInsurancePay	AVERAGE(FactTable[Insurance_Payment])	Executive Summary
%InsurancePay	[TotalInsurancePay] / [TotalPayment]	Executive Summary
%PaymentPay	[TotalPatientPay] / [TotalPayment]	Executive Summary
Insurance_A+	CALCULATE([TotalInsurancePay], BloodGroup = "A+")	Patient Analysis
Insurance_Tobacco	CALCULATE([TotalInsurancePay], Tobacco = "Yes")	Patient Analysis

7.2 Expense & Deficit Measures

Measure Name	DAX Formula	Used In
TotalGrossExpense	SUM(FactTable[Gross Expenses])	Exec Summary, Hospital Insights, Monthly Expenses
TotalDeficit	SUMX(FactTable, [Gross Expenses] - [TotalPayment])	Executive Summary
AverageDeficit	AVERAGEX(FactTable, [Gross Expenses] - [TotalPayment])	Executive Summary

BadDebts	CALCULATE([TotalGrossExpense], dimTransactionFK = 55711 || 48429 || 55042)	Hospital Insights

7.3 Time Intelligence Measures

Measure Name	DAX Formula	Used In
YTDPayment	CALCULATE([TotalPayment], DATESYTD(DimDate[Date]))	Executive Summary
QTDPayment	CALCULATE([TotalPayment], DATESQTD(DimDate[Date]))	Executive Summary
Previous MonthPay	CALCULATE([TotalPayment], DATEADD(DimDate[Date],-1,MONTH))	Executive Summary
TotalPayMA	DIVIDE(CALCULATE([TotalPayment], DATESINPERIOD(...,-10,DAY)),10)	Executive Summary
10DayRollingPay	CALCULATE([TotalPatientPay], DATESINPERIOD(...,-10,DAY))	Executive Summary

7.4 Efficiency Ratio Measures

Measure Name	DAX Formula	Used In
IPTP Ratio	SUM([Insurance_Payment]) / [TotalPayment]	Hospital Insights
ARGE Ratio	SUM(FactTable[AR]) / SUM(FactTable[Gross Expenses])	Hospital Insights

7.5 Volume & Demographic Measures

Measure Name	DAX Formula	Used In
DistinctPatient	DISTINCTCOUNT(FactTable[dimPatientFK])	Executive Summary, Patient Analysis
CountPatient	COUNT(Patient_Lookup[dimPatientPK])	Patient Analysis (base measure)
MalePatient	CALCULATE([CountPatient], PatientGender = "Male")	Patient Analysis
FemalePatient	CALCULATE([CountPatient], PatientGender = "Female")	Patient Analysis
TotalMales	CALCULATE([MalePatient], ALL(Patient_LookUp))	Patient Analysis (denominator)

TotalFemales	CALCULATE([FemalePatient], ALL(Patient_LookUp))	Patient Analysis (denominator)
%Males	DIVIDE([MalePatient],[TotalMales])	Patient Analysis
%Females	DIVIDE([FemalePatient],[TotalFemales])	Patient Analysis
Avg_Pay_Males_B+	CALCULATE([AveragePatientPay], Gender="Male", BloodGroup="B+")	Patient Analysis
Avg_Pay_Females_B+	CALCULATE([AveragePatientPay], Gender="Female", BloodGroup="B+")	Patient Analysis

8. Key Findings and Strategic Recommendations

8.1 Summary of Critical Findings

F1	NETWORK-WIDE DEFICIT: Total Gross Expenses ($13.0M) exceed Total Revenue ($9.0M) by $4.0M for the fiscal year - a fundamental financial imbalance that existed even before COVID-19 impact in 2020.
F2	COVID-19 DEMAND COLLAPSE: Patient volume fell 96% (4,800 → 200) and procedure volume fell 89% between January and December 2020. The collapse was demand-driven - providers were available but patients were not seeking care.
F3	OCTOBER EXPENSE ANOMALY: A 5.6x gross expense spike in October 2020 ($0.37M → $2.06M) with immediate reversal in November has no clear operational explanation and requires direct financial investigation.
F4	GENESIS PARADOX: Genesis Hospital Center has the best insurance collection efficiency (90% IPTP) but the worst revenue-to-expense ratio (19% ARGE) in the network - indicating a structural billing or cost issue.
F5	84K INACTIVE ENROLLEES: Only 5,117 of 84,000 enrolled patients (6%) are active. The 78,883 inactive enrollees represent an immediate, low-cost revenue expansion opportunity through re-engagement campaigns.
F6	NORTHEAST COVERAGE GAP: The Northeast has 30% fewer providers than the South (484 vs 690) despite being the most densely populated US region. This is either an unaddressed market gap or a patient leakage issue.
F7	SINGLE-PROVIDER SPECIALTY RISK: 12 specialties have only 1-2 providers. A single departure eliminates that clinical capability from the entire network with no backup coverage.
F8	PART-TIME CAPACITY OPPORTUNITY: 219 providers are part-time (headcount 928 vs FTE 709). Converting even 50% to full-time would add ~110 FTE at zero recruitment cost.

8.2 Strategic Recommendations

Immediate Actions (0-3 months)

- **Priority 1:** Investigate October 2020 expense spike - obtain transaction-level detail for the $2.06M gross expense figure and determine whether it represents a one-time event, deferred obligation, or reporting error.
- **Priority 1:** Audit Genesis Hospital Center billing practices - compare its CPT billing data against peer hospitals to identify the root cause of its 19% ARGE ratio despite 90% IPTP efficiency.
- **Priority 1:** Establish backup coverage agreements for the 12 single-provider specialties (Anesthesiology, Midwife, Plastic Surgery, Surgical Oncology, etc.) with regional health systems.

Short-Term Actions (3-6 months)

- **Priority 2:** Launch inactive patient re-engagement program targeting the 78,883 enrolled but inactive patients - even a 5% conversion rate would add approximately 3,944 active patients, increasing revenue by an estimated $1.5M-$2.5M.
- **Priority 2:** Develop a Northeast provider expansion strategy - conduct market analysis to determine whether the 206-provider gap reflects genuine demand or structural market differences before committing to hiring.

- **Priority 2:** Audit part-time provider contracts and identify which of the 219 part-time providers are willing and able to transition to full-time, prioritizing high-CPT-unit specialties (Radiology first at 438 CPT units per provider).

Strategic Initiatives (6-18 months)

- **Priority 3:** Rebalance NPP-to-Physician ratio from current 1:5 toward 1:3 through targeted NPP hiring in primary care specialties - projected to reduce labor costs by 10-20% while expanding appointment availability.
- **Priority 3:** Build patient lifestyle intervention programs targeting the sedentary + alcohol-consuming patient segment (estimated 37% of active patients) to reduce long-term chronic disease costs.
- **Priority 3:** Develop Radiology capacity expansion business case - Radiology's 438 CPT units per provider is the highest ROI in the network; a 20-provider expansion would generate approximately 8,760 additional CPT units annually.

8.3 Financial Recovery Outlook

The network entered fiscal year 2021 from an extremely depressed baseline: December 2020 gross expenses of $0.21M and patient counts of approximately 200 represent a floor, not a steady state. Recovery trajectory will depend on:

- Speed of patient re-engagement as COVID-19 conditions normalized in 2021
- Success of inactive enrollee activation (84K enrolled base)
- Insurer contract renegotiations affecting the IPTP and ARGE ratios
- Resolution of Genesis Hospital Center structural issues
- Execution of the Northeast provider expansion

Assuming a return to 50% of January 2020 activity levels, the network would generate approximately $4.5M in monthly gross expenses with $2.25M in adjusted payments - sufficient to break even at reduced operational scale pending further recovery.